



Data Driven Education

Discussion Forum Subteam

Presentation 1 - 02/15/2023

Team Members



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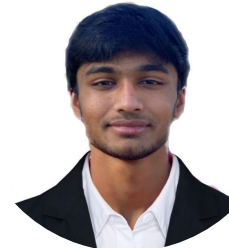
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Front-end
Web Developer



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Back-end
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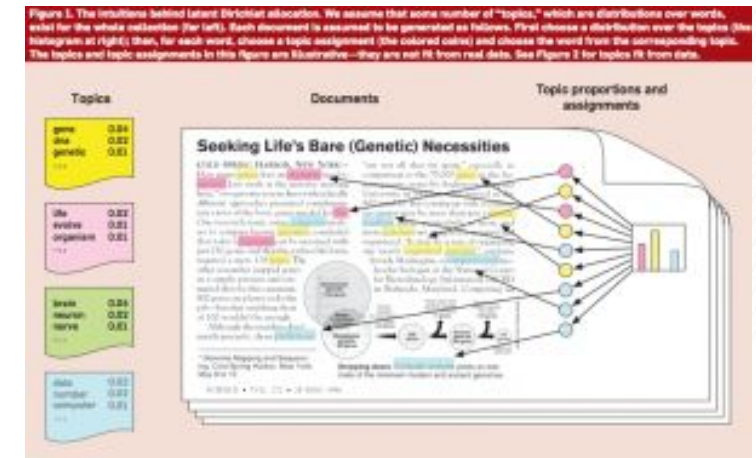
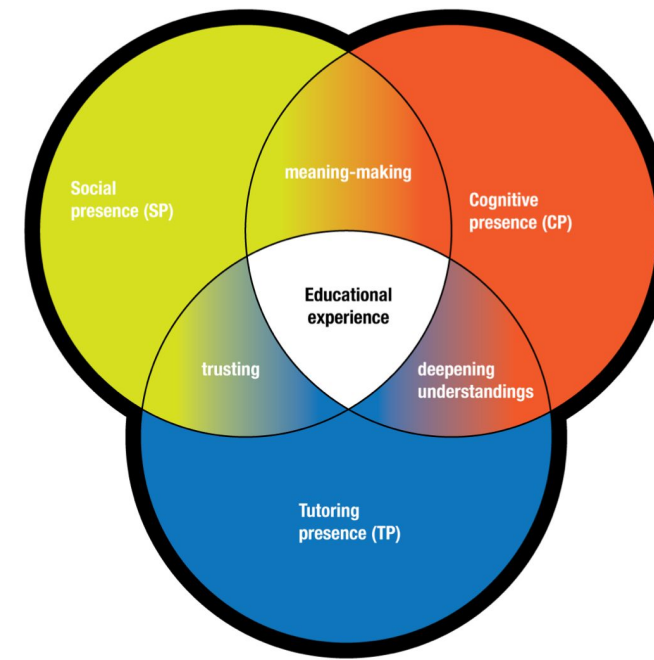


Rohan Karthik

Data Engineer

Introduction

- Discussion forums
 - main source of interaction in MOOCs
- Student interaction
 - Aspect-Based Sentiment Analysis model
 - Apply understandings from COI framework
 - Cognitive presence
 - Social presence
 - Topic Modeling



Goal: Further the applications of our dashboards to provide more accuracy and insights for our instructors

Recap of Fall 2022

Category	Accomplished	Next Goals
Web Dev	• A dashboard has been created with dummy data that shows what and how information will be presented to users.	• Develop two dashboards ○ Separate one for each class ○ incorporate more data points from the UX research of last semester ○ Utilize new UX research for our specific tools
UX Research	• Summary document with user needs for instructors based on thorough interviews	• Tailor additional interviews specifically to our outputs – ABSA and topic modeling
Data Science	• Sentiment Analysis model created, accuracy could have been higher	• Improve accuracy of sentiment analysis model • Develop topic modeling • Utilize the new metacognitive data

Goals as a Project Manager:

- Have clear deliverables each week for team members to make things concrete
- Be a resource for all members of the team
- Drive UX research with instructor interviews to determine user needs

Focus for this semester

● Development

- Two separate dashboards
 - (ML & CS 1301)
 - Sentiment Analysis
 - Topic Modeling
- Metacognitive data
 - Visualization

● Research

- Interview professors and students
 - Identify primary, secondary and tertiary needs
 - Focus interviews to the specific tools that we are developing
- Student participation
 - Calculate a participation score in Piazza (or on our dashboard), without having to export the data manually (CS 2340 TA, Dr. Joyner, Spring 2021)
 - Report students with the highest participation, based on CP, endorsed-answers, and activity
 - Top student answers and top student “instructor-endorsed” answers (CS 1332 TAs, Spring 2021 + general, Fall 2021)
 - Report a metric of overall participation in the class
 - Student sentiment
 - Overall sentiment of students (CS 2340 TA's, Spring 2021), although Vaibhav said that this largely stays the same semester to semester
 - Sentiment around a given HW or exam

● Project Management

- Integrating topic modeling and Aspect-Based sentiment analysis in the dashboards
- Bridging the gap between the Web-Development team and the Data Science team
- Research on user needs for this dashboard.
 - Supplement research done by previous subteam members by asking more questions with a specific focus on the tools that we are developing
 - Bring back focus to Community of Inquiry Framework

Research Questions

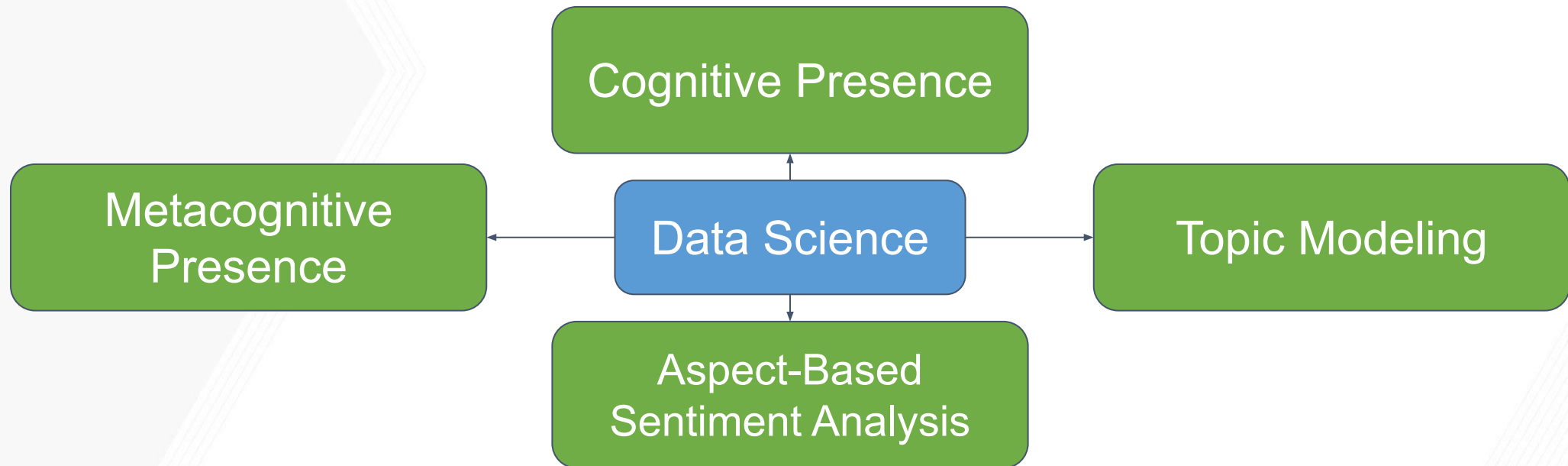


- Based on the community of inquiry framework, what are some useful metrics for student engagement and performance in an asynchronous online classroom?
 - how can we use the user feedback from last semester and expand our research to understand specific applications for the models we are developing?
 - how can we improve upon the accuracy of the models we've already developed?
- How can we design a web app that can provide actionable insights to the instructor?
 - how can we make the web app user-friendly and compatible with different devices?

Data Science

Data Science Big Picture

- Data available: CS1301 and CS6601
- Importance of Topic Modeling and Aspect-Based Sentiment Analysis (ABSA)



ABSA - Previous Work

- Aspect-Based Sentiment Analysis

- Technique that categorizes text data by aspect and identifies the sentiment attributed to each one
- Can provide reasonable sentiment for aspects depending on students' current circumstance for multiple aspects at once

- Tested and trained on Piazza data

- Example input:

- **Text:** for some reason from my end the ' gradescope ' submission is still not happening

- **Aspects:** [test, project, quiz, homework, gradescope, assignment]

- Example output:

```
Sentiment.negative for "gradescope"
```

```
Scores (neutral/negative/positive): [0.051 0.932 0.018]
```

```
Importance 1.00 for some reason from my end the ' gradescope ' submission is still not happening
```

```
Importance 0.51 for some reason from my end the ' gradescope ' submission is still not happening
```

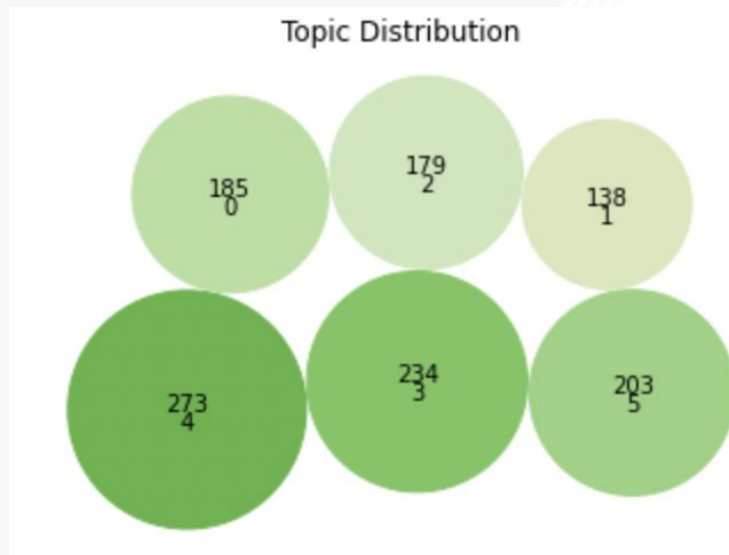
```
Importance 0.19 for some reason from my end the ' gradescope ' submission is still not happening
```

```
Importance 0.15 for some reason from my end the ' gradescope ' submission is still not happening
```

```
Importance 0.13 for some reason from my end the ' gradescope ' submission is still not happening
```

Topic Modeling - Previous Work

- Topic Modeling - unsupervised
 - scan documents, detect words, cluster groups
 - should be aware of dominant topic
 - limitation: the top topic was irrelevant to the course contents
- Visualization on Piazza data



Guided Topic Modeling - New Model

- Topic Modeling with predefined topics
 - setting several seed topics to which the model will converge
 - We provide seed words for each topic, so each topic word is related to specific contents

```
seed_topic_list = [  
    ['tree', 'decision', 'leaf', 'node'],  
    ['pycharm', 'version', 'python', 'numpy', 'scikit-learn', 'keras'],  
    ['gradescope', 'test', 'submission', 'quiz'],  
    ['confusion', 'matrix']  
]
```

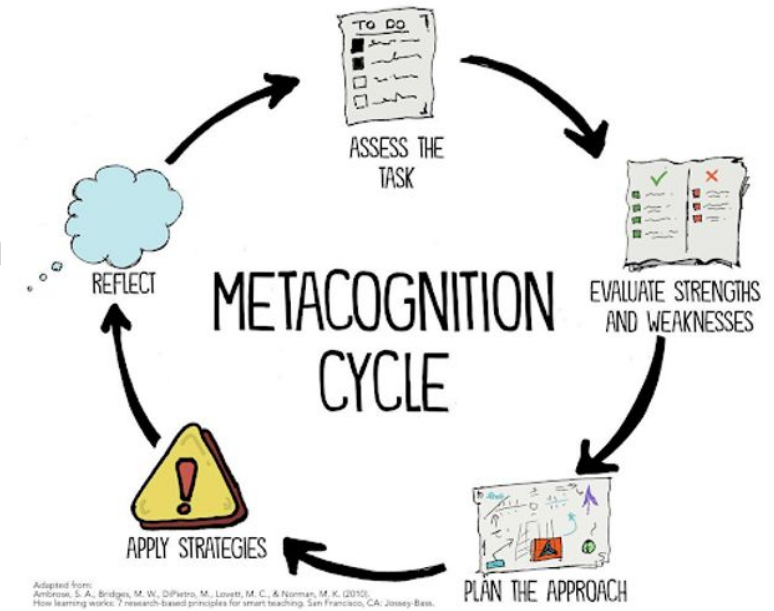
- Expectation
 - Reduction of running time by utilizing pipeline
 - Improvement of accuracy of classification by topic

Metacognitive Presence

Data - association between accuracy and confidence can be taken as a quantitative measure of metacognition

“Metacognitive practices help students become aware of their strengths and weaknesses as learners, writers, readers, test-takers, group members, etc.” – Bransford, Brown, & Cocking

- The model will ideally show the success of the student related to their metacognitive presence
- Will use posts that demonstrate metacognition and link this to success of the student in the class



Epoch 6: 100%

336/336 [0

Epoch 6

Training loss: 0.21516567942481266

Class: 0

Accuracy: 44/47

Class: 1

Accuracy: 0/4

Class: 2

Accuracy: 10/12

Class: 3

Accuracy: 2/5

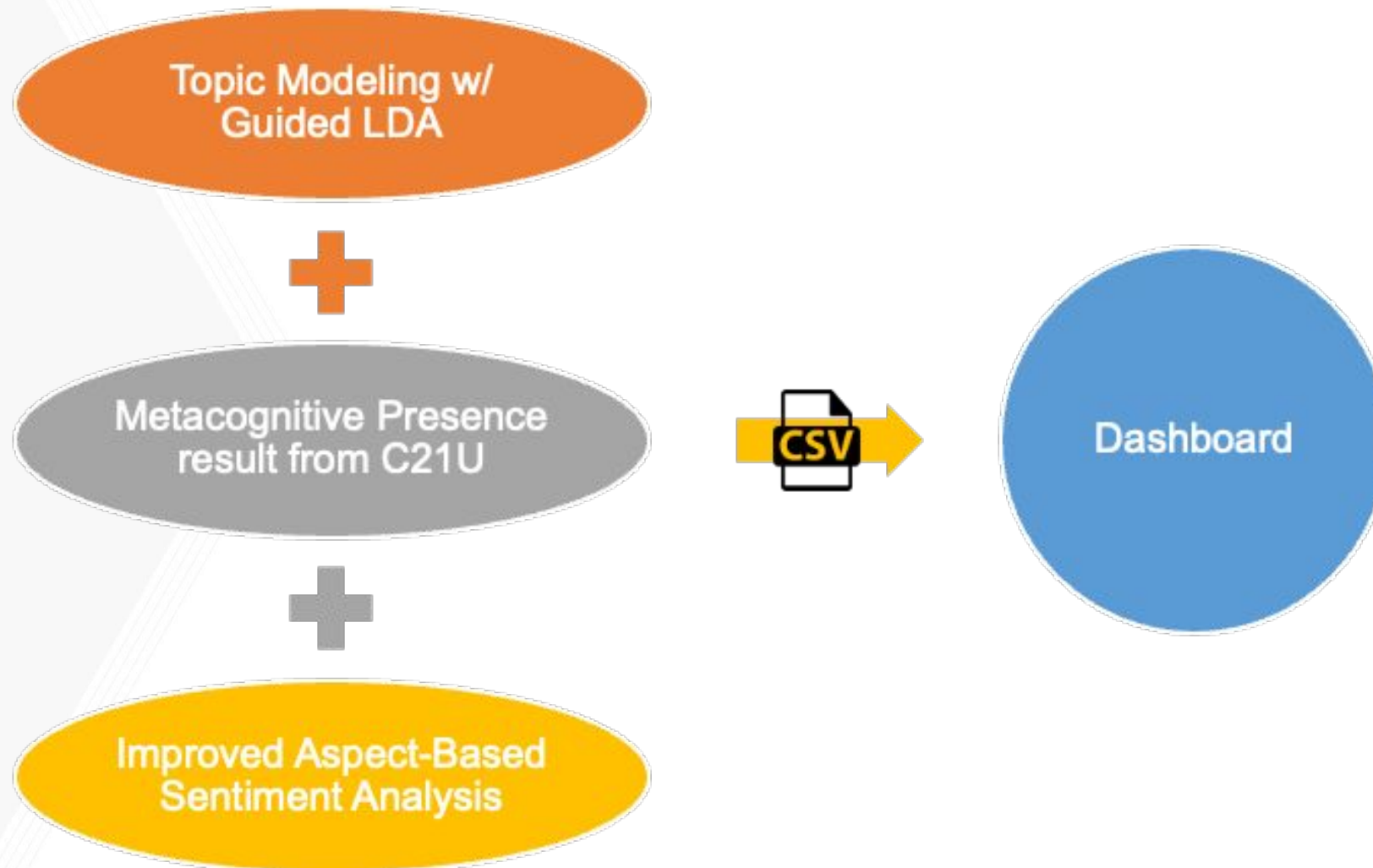
Class: 4

Accuracy: 5/6

Validation loss: 0.7220008489063108

F1 Score (Weighted): 0.8188340636794245

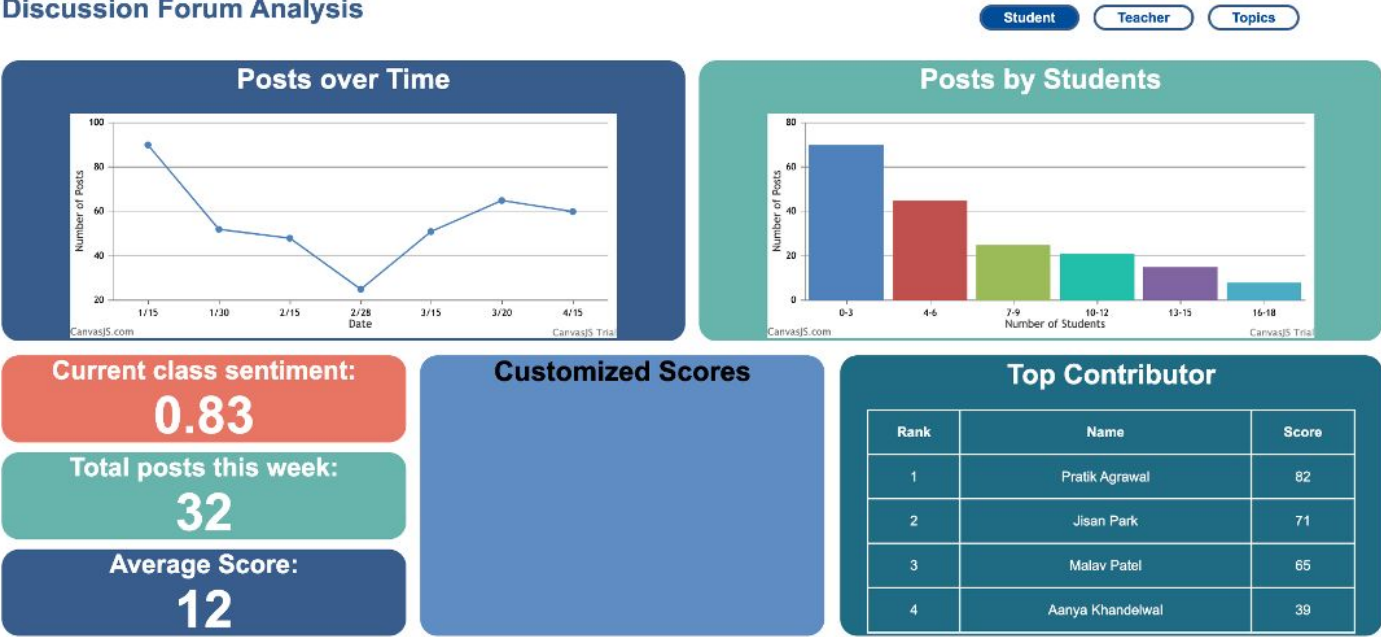
Overall Semester Goals



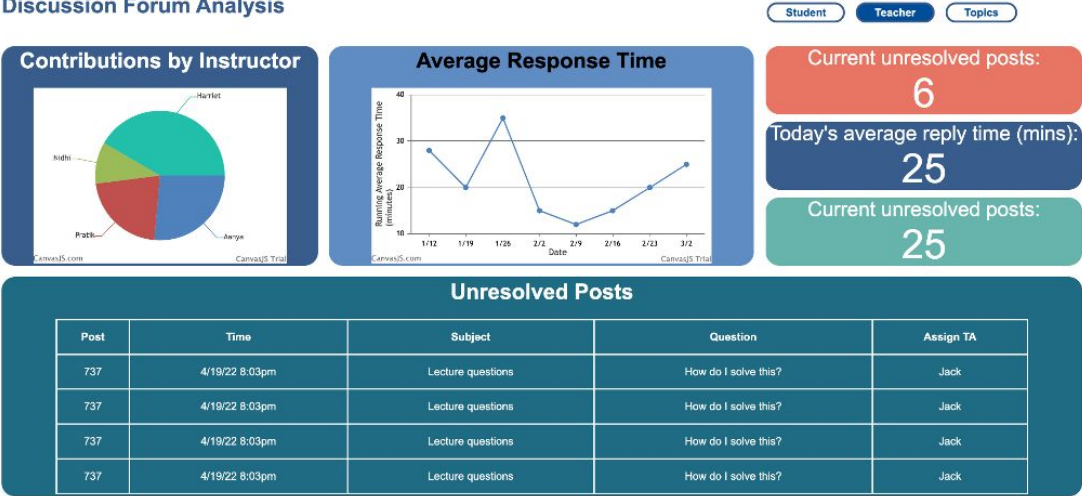
Web Development

Previous Work

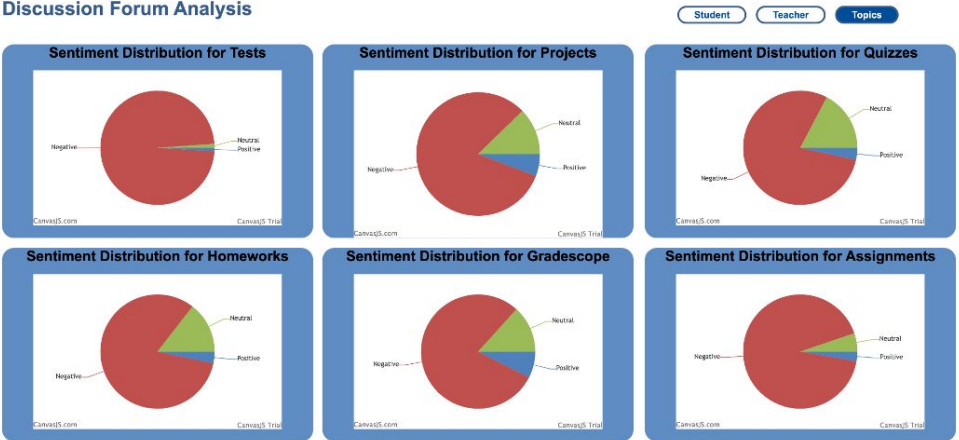
Discussion Forum Analysis



Discussion Forum Analysis



Discussion Forum Analysis



Goals for this semester

- Allow users to view data for multiple courses through the dashboards.
- Make the dashboard responsive so that it can be used on devices of all sizes.
- Interview with professors to find out the datas they find important and update the dashboard to show those information
 - Specialize interviews to focus on our tools for this semester, put into use previous interviews
- Connect the graphs, tables and all information in the dashboard to the backend data generated by the Data Science team.

Conclusion

- Two dashboards (CS6601 and CS1301)
 - Aspect-Based Sentiment Analysis - improved accuracy
 - Topic Modeling - improved design
- Utilize Metacognitive Coding results
- Web Development
 - Create second dashboard for the data from the additional class
 - Integrate user feedback from last semester
 - Expand user research to specific applications of above dashboards
 - Incorporate metacognitive coding results into the dashboard

Citations

1. Peacock, S., & Cowan, J. (2016). From Presences to Linked Influences Within Communities of Inquiry. *The International Review of Research in Open and Distributed Learning*, 17(5). <https://doi.org/10.19173/irrodl.v17i5.2602>
2. Mishra, Ankit, Sugasi Guatam, & Gehani Ritika. (2022). Web Development Team Report. *Georgia Tech Vertically Integrated Projects*.
3. Mcdaniel, Rhett. “Metacognition.” *Vanderbilt University*, Vanderbilt University, 9 Feb. 1970, <https://cft.vanderbilt.edu/guides-sub-pages/metacognition/>.
4. “Topic Modeling: A Basic Introduction.” *Journal of Digital Humanities*, <http://journalofdigitalhumanities.org/2-1/topic-modeling-a-basic-introduction-by-megan-r-brett/>.
5. Castellanos-Reyes, D. 20 Years of the Community of Inquiry Framework. *TechTrends* 64, 557–560 (2020). <https://doi.org/10.1007/s11528-020-00491-7>

Thanks for listening!

Questions?